Take-Home Exercise: Workflow Automation

4 hours · AI allowed · TypeScript on the MEAN stack

Overview

Build a small automation app inspired by Zapier's Zaps — not a Zapier clone. The must-have is signing in with GitHub and a Zap that comments on a pull request when one is opened. A natural-language Copilot that drafts Zaps is a bonus.

After you submit, we'll have a 30–40 minute follow-up about the app, your git history, and how you used AI.

Time

Four hours, on the honor system. Please note when you started and when you stopped.

How to work

Treat this like a real workday. Use the tools, AI, git workflow, branching, commit habits, and PR practice you actually use. We will look at the history and ask you about it. Please don't invent a take-home persona.

Tech stack

Use TypeScript throughout, on the MEAN stack:

- **MongoDB** for persistence
- **Express** on the server
- **Angular** for the front end
- **Node.js** as the runtime

Beyond that, libraries and tooling are your choice.

AI

You may use AI in any form. You are still the person we are hiring. In the follow-up, you should be able to walk through what you shipped, what you decided, and what you delegated.

What to build

A small workflow automation product. Take inspiration from Zapier's Zaps — a trigger plus an action — not the rest of Zapier (marketplace, thousands of apps, billing, teams). How close you stay to that metaphor is part of what we'll talk about.

The Zap that must work end to end

- People **sign in with GitHub** (OAuth / SSO).
- They can create a Zap: **GitHub pull request opened → comment on that same pull request**.
- Opening a real PR on a repo they control causes that comment to appear.

- Zaps belong to the signed-in user.
- The builder can list, create, and edit a Zap (name, on/off, trigger, action, and some field mapping).
- The app picker should include **several developer tools**. Only GitHub needs to actually run. What you do with the others is up to you.

GitHub access

Use your own GitHub account. We will not provide access to our organization.

Bonus

If the Zap above works: add a **Copilot** that builds a Zap from natural language into the same product, so a user could review it and turn it on. For example: *"When a pull request is opened, comment thanks."*

What to submit

A git repo we can run, with whatever README you would normally leave for a teammate.

The follow-up

We'll spend 30–40 minutes with you. We'll run the app, open a PR, walk through the code and the git history, and talk about how you used AI.

Good luck — we're looking forward to seeing what you build.